



Databases and Applications

Unit III

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Computer Applications

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Unit III

3NF, Transitive Dependencies, BCNF

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Computer Applications



Transitive Dependency – Definition

- ▶ **Transitive Dependency**
- ▶ A transitive dependency exists in a relation R if:
 - We have functional dependencies $X \rightarrow Y$ and $Y \rightarrow Z$
 - X is a key (or contains a key) of the relation
 - Z is a **non-key** attribute
- ▶ Then $X \rightarrow Z$ holds *indirectly* via Y, and we say Z is **transitively dependent** on X.
- ▶ 3NF aims to **remove** such transitive dependencies of non-key attributes on a candidate key.



Transitive Dependency – Employee Example

- ▶ Consider the **Employee** relation from the 3NF discussion:

EmpNum	EmpName	DeptNum	DeptName
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- ▶ Functional Dependencies:

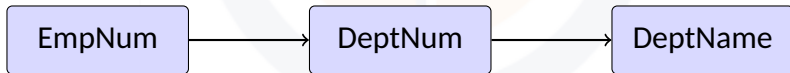
- $\text{EmpNum} \rightarrow \text{EmpName}$
- $\text{EmpNum} \rightarrow \text{DeptNum}$
- $\text{DeptNum} \rightarrow \text{DeptName}$

- ▶ The first two are direct dependencies from the candidate key EmpNum.



Transitive Dependency – Employee Example

- ▶ From the previous slide:
 - $\text{EmpNum} \rightarrow \text{DeptNum}$
 - $\text{DeptNum} \rightarrow \text{DeptName}$
- ▶ Therefore:
 - $\text{EmpNum} \rightarrow \text{DeptName}$ (indirect)
- ▶ This is a **transitive dependency** because DeptName is a non-key attribute depending on EmpNum *through* DeptNum.



- ▶ Because of this transitive dependency, the original Employee relation is **not in 3NF**.



Third Normal Form

- ▶ A relation is in **3NF** if the relation is in **1NF** and all determinants of non-key attributes are candidate keys.
 - That is, for any functional dependency: $X \rightarrow Y$, where Y is a non-key attribute (or a set of non-key attributes), X is a candidate key.
- ▶ This definition of 3NF differs from BCNF only in the specification of non-key attributes – 3NF is **weaker** than BCNF.
 - BCNF requires *all* determinants to be candidate keys.
- ▶ A relation in 3NF will not have any **transitive dependencies** of a non-key attribute on a candidate key through another non-key attribute.



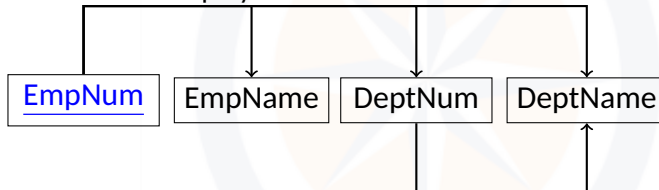
- ▶ Consider this **Employee** relation:

EmpNum	EmpName	DeptNum	DeptName
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- ▶ EmpName, DeptNum, and DeptName are **non-key attributes**.
- ▶ DeptNum determines DeptName, a non-key attribute, and DeptNum is not a candidate key.
- ▶ From the slide:
 - Is the relation in 3NF? ... **no**
 - Is the relation in BCNF? ... **no**
 - Is the relation in 2NF? ... **yes**
 - Candidate keys are? ... **EmpNum**



- ▶ Consider the Employee relation:





► Functional dependencies:

- $\text{EmpNum} \rightarrow \text{EmpName}$
- $\text{EmpNum} \rightarrow \text{DeptNum}$
- $\text{DeptNum} \rightarrow \text{DeptName}$

► This creates a **transitive dependency**:

$\text{EmpNum} \rightarrow \text{DeptNum} \rightarrow \text{DeptName}$

- This violates 3NF because a non-key attribute depends on another non-key attribute.



- ▶ We correct the transitive dependency by decomposing the relation into 3NF relations:

- ▶ Relation 1

EmpNum	EmpName	DeptNum
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- ▶ Relation 2

DeptNum	DeptName
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- ▶ Both resulting relations satisfy 3NF.



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3NF but NOT BCNF

- ▶ Example: *Instructor teaches one course only; student takes a course and has one instructor.*

student_no	course_no	instr_no
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- ▶ Functional dependencies:
 - student_no, course_no $\rightarrow$ instr_no
 - instr_no $\rightarrow$ course_no
- ▶ Since instr_no is not a candidate key, the relation is NOT in BCNF.



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Boyce-Codd Normal Form (BCNF)

► A relation is in **BCNF** if:

- It is already in 3NF
- For every functional dependency $X \rightarrow Y$:
 - The determinant X is a **superkey (SK)**

► **Why BCNF?**

- 3NF still allows some anomalies.
- Non-superkey determinants cause redundancy.
- BCNF ensures all dependencies arise from keys only.

► **Characteristics**

- Every FD satisfies: LHS is a superkey.
- BCNF is stricter than 3NF.
- Removes all update, insertion, and deletion anomalies.

► **Observation**

- If a relation has only one candidate key, 3NF = BCNF.



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Boyce-Codd Normal Form (BCNF)

► Common BCNF Violation Example

- Relation: *Teach*(*Course*, *Instructor*, *Room*)
- FDs:

$Course \rightarrow Instructor, \quad Instructor \rightarrow Room$

- Problem:
 - *Instructor* is not a superkey
- ⇒ Relation is in 3NF but **not in BCNF**.

► BCNF Decomposition Rule

- If $X \rightarrow Y$ violates BCNF (i.e., X not SK), decompose into:
 - $R_1 = X \cup Y$
 - $R_2 = R - (Y - X)$
- Decomposition is always **lossless**.
- Further decomposition may be needed until all relations satisfy BCNF.



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BCNF – Decomposition

- ▶ We decompose the relation into BCNF relations:

student_no	instr_no
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course_no	instr_no
-----------	----------

- ▶ The decomposition results in BCNF relations.



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Normalisation – Visual Summary

1NF – Atomic values

2NF – No partial dependencies

3NF – No transitive dependencies

BCNF – LHS of every FD is a SK



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Summary of Normalisation

- ▶ Normalisation improves database design by reducing redundancy and eliminating anomalies.
- ▶ **1NF – First Normal Form**
 - No repeating groups or multivalued attributes
 - All attribute values are atomic
- ▶ **2NF – Second Normal Form**
 - Must be in 1NF
 - No partial dependency on a composite key
 - Every non-key attribute depends on the *whole* key



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Summary of Normalisation

► 3NF – Third Normal Form

- Must be in 2NF
- No transitive dependencies on key attributes
- All determinants must be candidate keys, except when the dependent attribute is a prime attribute

► BCNF – Boyce-Codd Normal Form

- Must be in 3NF
- For every functional dependency $X \rightarrow Y$, X must be a superkey
- Stronger than 3NF; removes remaining anomalies



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Normalisation – Practice Question 1

► Consider a relation

$$R = (A, B, C, D, E)$$

with functional dependencies:

$$\{ CE \rightarrow D, D \rightarrow B, C \rightarrow A \}$$

► Tasks:

- Find all candidate keys of R .
- Determine the highest normal form satisfied by R (1NF, 2NF, 3NF).
- If the relation is not in 3NF, decompose it until it becomes 3NF.

At each decomposition step, form a new relation, recompute candidate keys, and identify its normal form.



► Given:

Relation: $R(A, B, C, D, E)$

FDs: $\{CE \rightarrow D, D \rightarrow B, C \rightarrow A\}$

► Candidate Keys

- Attributes that never appear on RHS: C, E
- Check $(CE)^+ = \{A, B, C, D, E\} \Rightarrow CE$ is a key.
- No proper subset of CE is a key $\Rightarrow$ **only candidate key: CE .**

► Highest Normal Form of R

- Partial dependency: $C \rightarrow A$ (non-prime A depends on part of key C).
- Transitive dependency: $CE \rightarrow D \rightarrow B$.
- Hence R violates 2NF and 3NF.
 $\Rightarrow$ **Best normal form of R : 1NF.**



► 3NF Decomposition

- From minimal cover, create relations:

$R_1(C, E, D)$ from $CE \rightarrow D$

$R_2(D, B)$ from $D \rightarrow B$

$R_3(C, A)$ from $C \rightarrow A$

- R_1 contains key CE for the original schema.
- In each relation, the determinant is a key $\Rightarrow$ all R_1, R_2, R_3 are in 3NF.



► Consider a relation

$$R = (A, B, C, D, E, F, G, H, I, J)$$

with the following functional dependencies:

$$\{ AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ \}$$

► Tasks:

- Determine all candidate keys of R .
- Identify the highest normal form satisfied by R (1NF, 2NF, 3NF).
- If the relation is not in 3NF, decompose it step-by-step until it reaches 3NF.

During each step, form a new relation, recompute its candidate keys, and check the normal form it satisfies.



► Given:

Relation: $R(A, B, C, D, E, F, G, H, I, J)$

FDs:

$$\{ AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ \}$$

► Candidate Keys

- Attributes never on RHS: $A, B \Rightarrow$ must appear in every key.
- Compute closure:

$$(AB)^+ = \{A, B, C, D, E, F, G, H, I, J\}$$

- Neither A^+ nor B^+ gives all attributes.
 $\Rightarrow$ **Only candidate key: AB.**



► Highest Normal Form

- Partial dependencies:

$A \rightarrow DE$ (only part of key)

$B \rightarrow F$ (only part of key)

- Transitive dependency:

$AB \rightarrow D \rightarrow IJ$

⇒ **R violates 2NF and 3NF.**

⇒ **Best normal form of R = 1NF.**



► 3NF Decomposition

- Using minimal cover, generate relations:

$R_1(A, B, C)$ from $AB \rightarrow C$

$R_2(A, D, E)$ from $A \rightarrow DE$

$R_3(B, F)$ from $B \rightarrow F$

$R_4(F, G, H)$ from $F \rightarrow GH$

$R_5(D, I, J)$ from $D \rightarrow IJ$

- R_1 contains the candidate key AB of the original relation.
- In every R_i , the determinant is a key $\Rightarrow$ each relation is in **3NF**.

Final decomposition:

$$\{R_1, R_2, R_3, R_4, R_5\}$$



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CELEBRATING 50 YEARS

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